NILAY SINGH

Jaipur, Rajasthan

+91 7014557020 | nilay109singh@gmail.com | LinkedIn | GitHub | HackerRank

EDUCATION

Vellore Institute of Technology — **September 2022 – August 2026**

Btech in Computer Science and Engineering

TECHNICAL SKILLS

Languages: Python, SQL, HTML, CSS , Java

ML/AI Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Developer Tools: Jupiter Notebook, Google Collaboratory, VS Code, Python IDLE, MySQL Workbench, MySQL Command Line Client, MS Excel, Power BI, CSV

Databases: MySQL

Core Concepts: Database Management, Exploratory Data Analysis (EDA), Machine Learning, Supervised Learning, Classification, Regression, Data Preprocessing, Model Evaluation

PROJECTS

Student Score Prediction | Machine Learning, Python, Linear Regression, Pandas, Matplotlib

- Built a machine learning model using Linear Regression to predict student scores based on study hours.
- Created and processed a structured dataset, performing basic data handling using **Pandas.**
- Trained the model to learn the relationship between input (study hours) and output (scores).
- Generated predictions for unseen data and visualized results using Matplotlib.
- Demonstrated understanding of supervised learning and regression concepts through a real-world example.

Smart Transaction Monitoring System | Machine Learning, Python, Scikit-learn, Pandas, Matplotlib

- Developed a machine learning-based fraud detection system using Random Forest to classify transactions as fraudulent or normal.
- Trained the model using features such as transaction amount and transaction frequency to identify suspicious patterns.
- Performed data preprocessing and feature selection using Pandas for efficient model training.
- Visualized transaction patterns and fraud distribution using Matplotlib to gain insights into data behavior.
- Implemented prediction logic to simulate real-time fraud detection

AirBnB Booking Analysis | Python, Pandas, Matplotlib, Seaborn, EDA, Data Visualization

- Conducted exploratory data analysis on **Airbnb listings dataset(102599 listings)** to uncover pricing patterns and customer booking trends using **Python**.
- Performed data cleaning, handling missing values, and feature analysis using **Pandas and NumPy.**
- Built data visualizations (heatmaps, distribution plots, correlation analysis) to identify relationships between price, location, room type, and availability.
- Generated actionable insights that could help optimize listing pricing and improve booking performance. Answered **5+ Analytial Questions** and created **8+ Visualizations**

Trainfo | Python, MySQL

- Designed and developed a desktop-based ticket reservation system that enables users to book tickets, view reservations, check train routes, and search available trains through an intuitive GUI.
- Implemented a **MySQL** relational database to manage trains, routes, users, and booking records, ensuring data integrity and efficient query performance.
- Built the application backend using **Python and SQL**, integrating **mysql.connector, pymysql, and SQLAlchemy** for secure and efficient database connectivity.
- Utilized **Pandas** for data processing, validation, and transformation of train schedules and reservation data.
- Created a user-friendly interface using **EasyGUI**, allowing non-technical users to interact with the system easily.

CERTIFICATIONS AND ACHIEVEMENTS

- HackerRank- SQL (Intermediate) Certificate
- Python Essentials - Vityarthi
- 4 Star SQL - HackerRank
- HackerRank- SQL (Basic) Certificate
- Cloud Computing - NPTEL
- The Bits and Bytes of Computer Networking - Coursera
- Solved 20+ SQL/database problems on **LeetCode**, improving data querying and problem-solving skills.